

CAPSTONE PROJECT

A Latent Profile Analysis of Attachment Dimensions

Their Associations with Sleep, Mood, and Fatigue

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Background & Rationale

- Attachment theory posits that early relational experiences shape internal working models governing interpersonal behavior and emotion regulation (Adams et al., 2014).
- Insecure attachment may disrupt sleep through heightened vigilance, emotional dysregulation, and pre-sleep arousal (Hu et al., 2024).
- Mixed findings in adult attachment–sleep research highlight the need to better capture heterogeneity in attachment patterns across adulthood (Hu et al., 2024; Xie et al., 2022).
- A person-centered approach (LPA) can identify distinct attachment profiles and examine how these profiles relate to wellbeing outcomes.

Why LPA?

- Captures co-occurring patterns of attachment anxiety + avoidance
- Identifies unobserved attachment subgroups
- Moves beyond average effects of single variables
- Links attachment profiles to sleep and wellbeing outcomes

Research Questions & Hypotheses

RQ1

What distinct attachment profiles emerge from dimensions of attachment anxiety and avoidance?

H: I hypothesize 3–4 profiles (e.g., Secure, Anxious, Avoidant, Fearful).

RQ2

Do profiles differ in self-reported sleep quality?

H: Secure profiles will report better sleep;
Anxious profiles will show the poorest sleep.

RQ3

Do profiles differ in mood (anxious/depressed)?

H: Fearful/Anxious profiles will have higher anxiety and depression.

RQ4

Do profiles differ in fatigue levels?

H: Higher anxiety and avoidance profiles will report greater fatigue.

Method: Participants & Design

Sample Size

N = 220

Adults in a relationship (≥ 3 months), English-speaking

Age Range

18–62

Mean age = 32.35 years

Design

**Cross-
Sectional**

Online survey study

Measures

ECR-S

Experiences in Close Relationships-Short (12 items) — attachment anxiety & avoidance

PHQ4

Patient Health Questionnaire (4 items) — anxiety & depression

PSQI

Pittsburgh Sleep Quality Index — overall sleep quality

MDBF

Multidimensional Mood Questionnaire fatigue item — daily fatigue

LPA Model Fit & Profile Selection

Model Fit Indices by Number of Profiles

Profiles	AIC	BIC	Entropy	n_min	
1	7527.97	7609.42	1.00	1.00	
2	7070.27	7195.84	.90	.32	
3	6977.17	7146.85	.84	.26	
4 ✓	6910.14	7123.93	.84	.13	
5	6877.22	7135.14	.83	.10	
6	6817.53	7119.57	.83	.05	

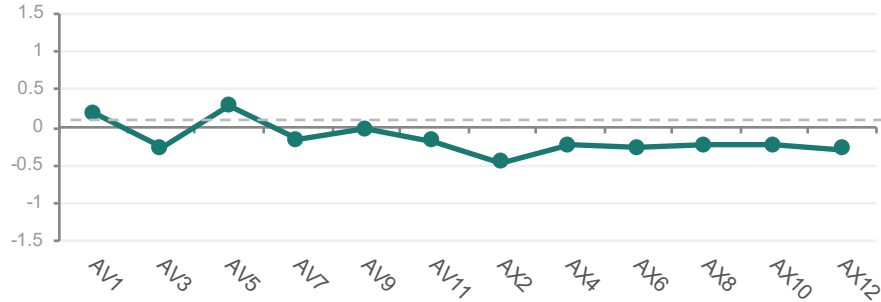
✓ Selected 4-profile solution

Selection Rationale

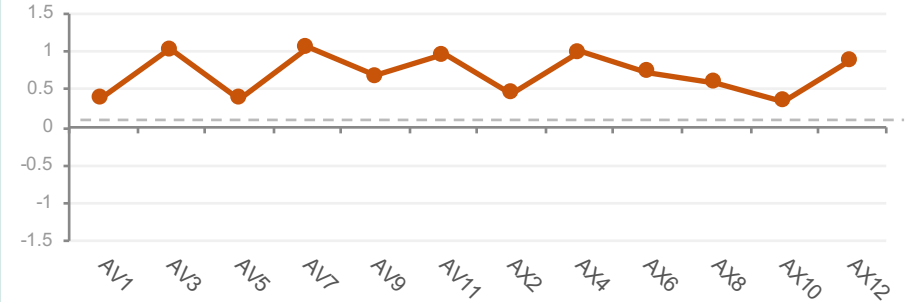
- Entropy remained acceptable for the 4-profile solution (.84)
- BIC worsened at 5 profiles
- Theoretically interpretable profiles
- Smallest class n > 5% of sample

Attachment Profile Plot

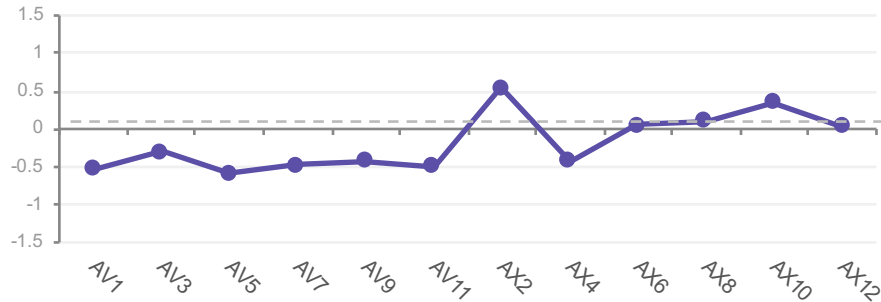
Profile 1: Dismissive-leaning



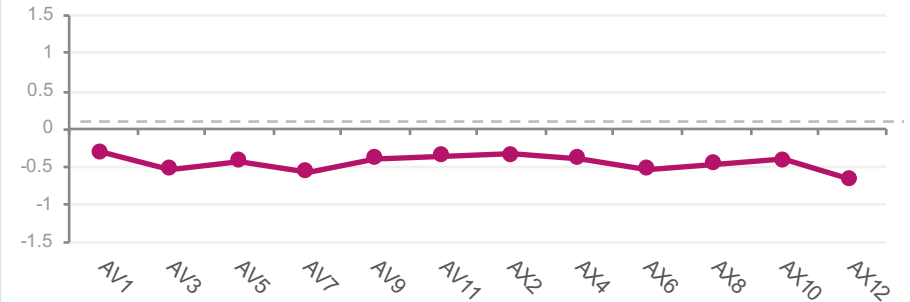
Profile 2: Fearful-avoidant



Profile 3: Anxious-preoccupied



Profile 4: Secure



AV1=TurnNeed | AV3=PullBack | AV5=Comfort | AV7=AvoidClose | AV9=Discuss | AV11=NervousClose
AX2=Reassure | AX4=NotClose | AX6=ScareAway | AX8=NoAbandon | AX10=Frustrated | AX12=WorryCare

Y-axis -> standardized mean score

Demographic Characteristics by Profile

Values are % for categorical variables and M (SD) for continuous variables. Highlighted cells indicate noteworthy patterns.

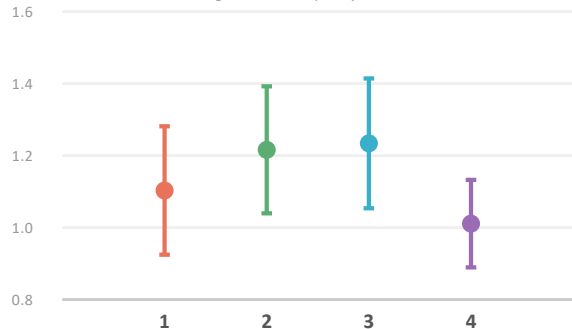
Variable	P1: Dismissive-leaning (n=29)	P2: Fearful (n=51)	P3: Preoccupied (n=47)	P4: Secure (n=93)
Age, M (SD)	35.7 (10.3)	31.7 (8.2)	30.5 (8.1)	32.6 (9.1)
Gender: Female %	27.6%	58.8%	72.3%	51.6%
Ethnicity: White %	65.5%	35.3%	34%	58.1%
Ethnicity: Black, Caribbean, or African %	24.1%	47.1%	53.2%	38.7%
Living with spouse %	82.8%	74.5%	76.6%	80.6%
Relationship > 3 years %	68.9%	62.7%	66%	69.9%

 = Notable pattern worth discussing

Sleep Outcomes by Attachment Profile

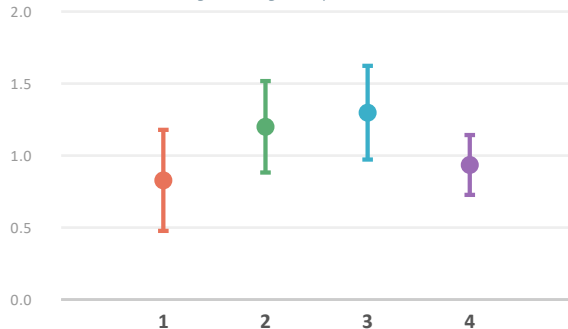
Sleep Quality

higher = worse quality



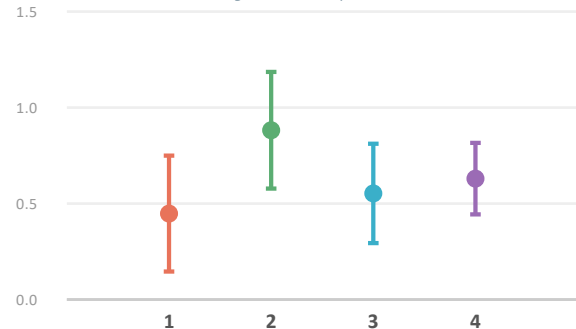
Sleep Onset

higher = longer sleep onset



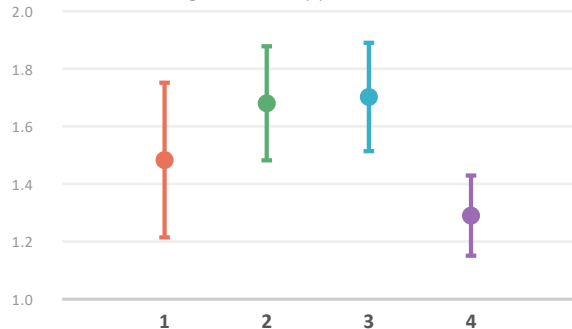
Sleep Duration

higher = less sleep



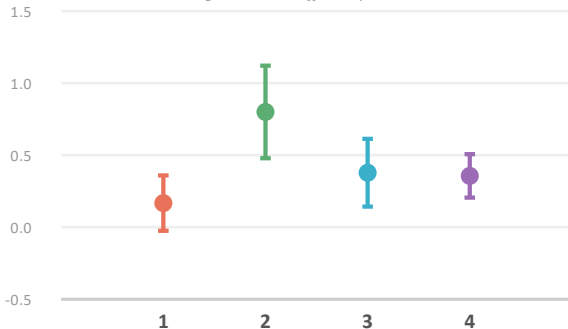
Sleep Problems

higher = more sleep problems



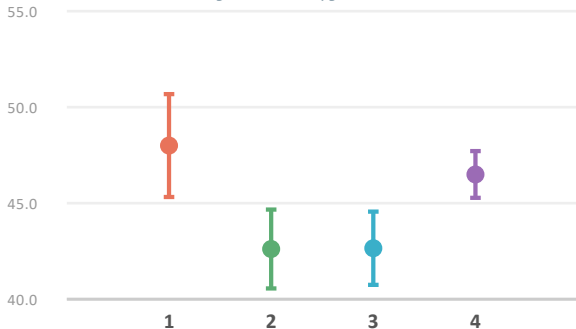
Sleep Efficiency

higher = worse efficiency



Sleep Hygiene

higher = better hygiene



Attachment Profile



P1: Dismissive



P2: Fearful

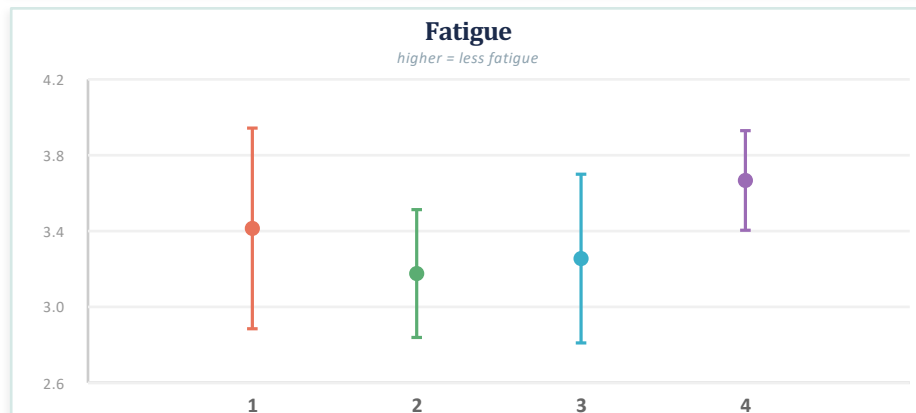
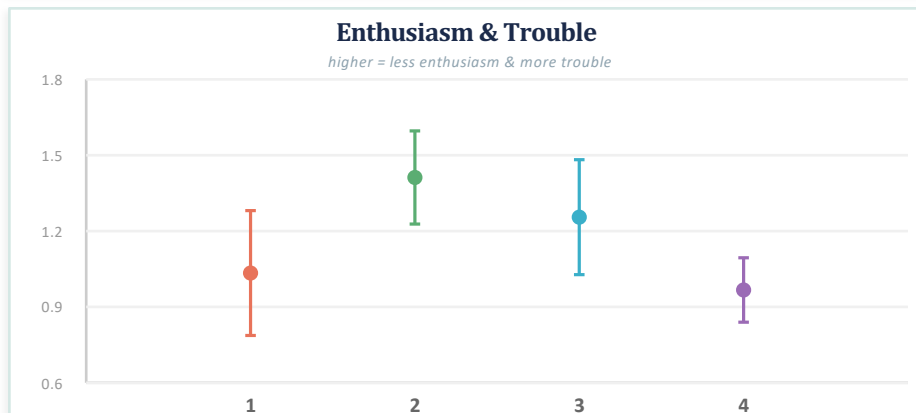
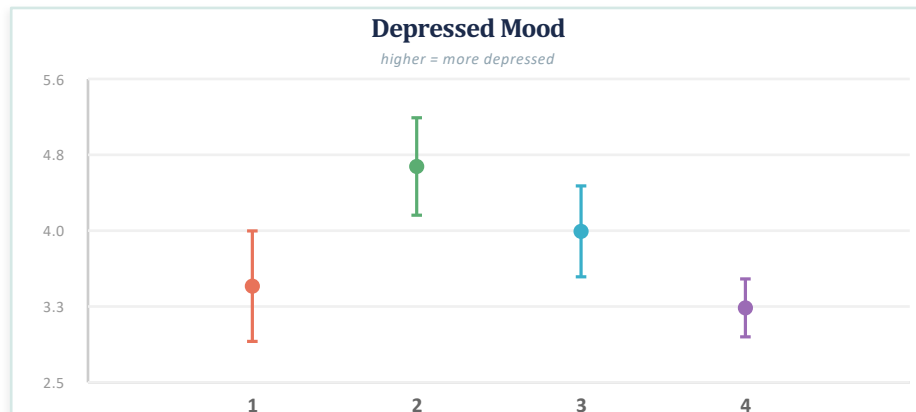
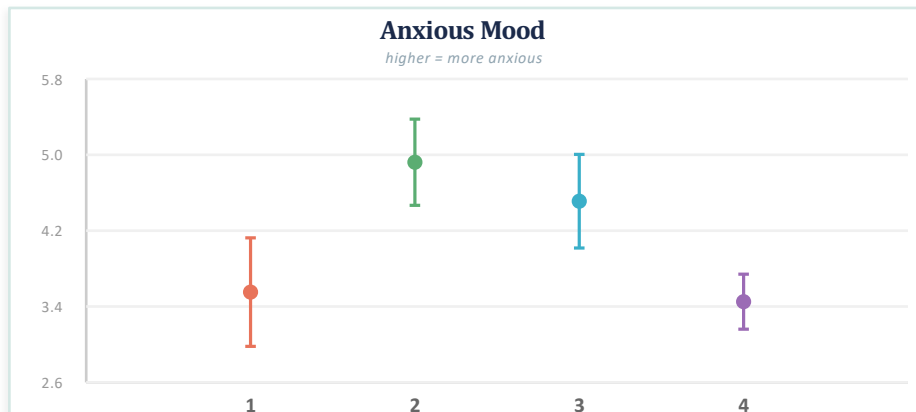


P3: Preoccupied



P4: Secure

Mood & Fatigue by Attachment Profile



● P1: Dismissive

● P2: Fearful

Attachment Profile

● P3: Preoccupied

● P4: Secure

Discussion

Profile Replication

Four-profile solution matched the classic four attachment styles (Secure, Preoccupied, Dismissing, Fearful), consistent with Bartholomew & Horowitz (1991).

Fearful Profile Most At-Risk

Fearful individuals (high on both dimensions) showed worst outcomes across nearly all wellbeing indicators.

Preoccupied — Reassurance-Seeking

Preoccupied profile was predominantly female and highest in reassurance-seeking, along with poorest sleep efficiency.

Dismissing — A Surprising Resilience

Dismissing individuals performed comparably to the Secure group across sleep and mood, with the best sleep hygiene.

Person-Centered Added Value

LPA revealed heterogeneity invisible in variable-centered analyses; anxiety, not avoidance alone, is the key driver of poor sleep and mood outcomes.

Limitations & Future Directions

Limitations

- Cross-sectional design limits causal inference
- Self-report measures subject to response bias
- Small sample size
- No definitive rule for choosing profile number
- Profile assignment is probabilistic, not certain

→ Future Directions

- Longitudinal designs to examine profile stability and health change
- Include objective sleep measures (actigraphy, polysomnography)
- Replicate with larger and more diverse community samples
- Cross-validate profile number selection in independent samples
- Incorporate classification uncertainty into future outcome analyses